



Implications of Artificial Intelligence on Digital Marketing and Providing Value to Customers in Bangladesh Perspectives

Mohammad Samin Ihsan*

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ABSTRACT

Businesses are leveraging Artificial Intelligence (AI) to create a better understanding of their consumers, craft marketing plans, and develop products and services. The internet connectivity, social media adoption, mobile financial services and digital commerce are growing at a rapid pace and have paved a special atmosphere for digital marketing with the help of AI in Bangladesh. The study provides a snapshot of the use of machine learning, natural language processing, and predictive analytics to provide personalized consumer experience and how technologies such as these can be leveraged by businesses for the best possible return on engagement marketing investments. This research findings the role of AI in digital marketing strategies and its value proposition for consumers. In a world where the digital landscape is constantly changing, companies need to evolve their marketing methods to meet consumers in the ways they are using now. The probe uses a mixed-method research approach, which comprises a quantitative examination in the form of descriptive statistics and a quantitative examination in the form of correlation analysis in addition to regression analysis. The study reveals strong linkages between the use of AI marketing strategies and important customer engagement, satisfaction and loyalty metrics. The findings offer a complex and nuanced view of the opportunities and risks AI brings about and help companies make well-informed decisions regarding their marketing investments in today's digital world. This study concludes that enterprises who use AI will get ahead in the race and those who do not and businesses must carve out a niche by offering a unique experience for their customers and ensuring sustainable growth through savvy use of AI.

1. Introduction

Digital marketing is no exception to the revolution that artificial intelligence (AI) has brought, transforming every aspect of the customer journey and the ways in which value is delivered. Businesses have access to vast amounts of data, thanks to the advanced capabilities of AI tools, such as machine learning, natural language processing (NLP), and predictive analytics, which are capable of analyzing consumer preferences or behaviour or even industry trends. That function has led to unimaginable magnitudes of precision and productivity in how fashion companies utilize their marketing tactics, thus AI has become a crucial instrument for reaching the maximum potential when advertising with today's technology (Chaffey 2023).

With the advent of AI, marketers are better equipped to forecast leads and optimize their content in a personalized manner as well as set up organizational tools for email campaigns or scheduling blog posts which has become more challenging due to complex user behavior this means less structure. Marketers can deploy these functions for better messaging, resulting in immediate campaign outcomes and stronger affinity with the customer (Kumar et al., 2021). Through AI, marketers can adjust their strategies to the changing needs of customers faster and with more agility.

AI has demonstrated an ability to bring customers more value and the same concept can potentially enable businesses to begin connecting these super-personalized recommendations for products, offers, or services that come up with individual preferences and behaviors. AI-powered chatbots and virtual assistants are changing the face of consumer

* Researcher, Dhaka, E-mail:

service for good, providing immediate responses 24/7 to instantly provide relief to customers by resolving any queries or problems. Those advancements result in a more seamless, convenient, and higher quality of consumer interaction reinforcing the loyalty towards your brand. For instance, AI's predictive power about what customers want to experience in different services and goods allows businesses to provide products that fully correspond with the tastes of a unique single person (Wirtz & Zeithaml 2022), significantly increasing customer retention rates while also improving Customer Lifetime Value (CLV).

However, with the advantages comes a range of threats that come with relying on AI in digital marketing. Collecting and analyzing huge data sets on the scale that most analytics operations occur today could also put collected information at risk of being compromised with unauthorized access to these vast caches, effectively eroding consumers' trust. In addition to these, serious matters of ethics must be considered, particularly regarding algorithmic bias and how much algorithmic input is being used for decision-making processes. These worries underscore the need to design, deploy and train AI systems responsibly and transparently while developing an ethical framework to maintain the responsibility of human power with regard to how society values care provision.

It's significant to grasp how AI remodels the form of existing marketing strategies and client experiences. In today's world, data, technology and automation are the backbone of the digital marketing ecosystem, and as a result, there are several ways AI can enhance business strategies to maximize their efficiency and effectiveness when it comes to customer satisfaction, retention and loyalty. It will also provide an all-in-one resource to delve into opportunities and challenges regarding AI in digital marketing and learn strategies for managing data privacy concerns. The findings from these studies can help companies and marketers better understand the potential of AI in their marketing strategies and how it can be applied more effectively, as well as offer insights into the use of AI for customer engagement and business innovation. This paper will review the ways in which AI is reshaping customer value creation and marketing plans, enabling businesses to adapt to the dynamics of the digital era.

2. Problem Statement

Digital transformation has brought new opportunities to businesses in Bangladesh, but it has also made it harder for businesses to capture customer's attention. Digital transformation has generated new opportunities for the businesses in Bangladesh but, on the other hand, it has also made it difficult for businesses to get the attention of the customers. The customers are able to check various price, product, service and reviews from different digital platforms. Social-media advertising enables businesses to reach large audiences, e-commerce and mobile financial services have some of the traditional geographical barriers removed (Huang and Rust, 2021). In such a context, mass marketing might not be enough. With the growing demand for personalised communication, businesses need to keep up with the real-time customer support and data-driven decision-making. AI equips technologies that can help meet these needs. However, there are still some issues to be addressed. AI tools in digital marketing could help optimize the management of marketing communication but won't necessarily build strong customer relationships. Over-automation can lead to a lack of human touch in customer interactions. Customer data is the backbone of AI-driven marketing. The gathering, processing and profiling of customer data raise issues related to privacy, transparency and cyber security. The concerns are especially relevant in the context of Bangladesh's evolving AI regulation and governance framework. Not all of your customers can access the advantages of digital marketing. The country is already making significant progress in terms of internet and mobile connectivity, but much of the population are still out of the web. By the end of 2025, DataReportal estimated that 53% of the population was still offline (DataReportal, 2025). Lastly, there is relatively little empirical literature that explores the relationship of AI-driven digital marketing with customer engagement, satisfaction, loyalty and perceived value in Bangladesh. The current international literature on AI in marketing is conceptual or global in scope. This study thus investigates whether AI-mediated digital marketing has the same impact on various dimensions of customer value, which is considered a problem in the context of Bangladesh.

3. Rationale of the Study

AI has transformed how businesses connect with customers and interact with them in a meaningful way, making digital marketing an essential component of any business. Until now, marketers played on audiences and spun original stories for each individual asset manually. However, when consumers' behavior changed, these became increasingly inefficient. With the arrival of AI, however, that has been eliminated, and new innovative technologies are introduced like learning algorithms, Natural Language Processing. The three key areas in focus are NLP, and predictive analytics. With tools like this, businesses can process vast amounts of data, identify patterns, and deliver truly customised customer experiences.

AI-Based Marketing has brought in another value to engaging customers and managing campaigns. Likewise, with technology, marketers have adopted recommendation systems and chatbots that utilize complex algorithms to deliver more personalized interactions to customers, and to automate the more exacting tasks. AI has taken the personalization trend one step further by recommending the most relevant products to customers and enabling real-time customer engagement, thus guaranteeing satisfaction and loyalty. The rise in the use of AI brings with it concerns regarding data privacy, potential for biased algorithms, and ethical use. Despite the power of AI, marketers will need to do everything in their power to stay within ethical boundaries and maintain customer trust by using data responsibly, with human oversight.

This study will examine how AI is impacting digital marketing strategies and customer value creation. The study will analyze the impact of AI on digital marketing strategies and customer value creation. It's a good idea for any company navigating this increasingly complex digital landscape to make sense of AI as a disruptor, providing insights into how to improve marketing strategies and how to deal with ethical and data privacy concerns.

4. Literature Review and Hypothesis Development

4.1. Artificial Intelligence Technologies in Marketing

AI's capacity to process vast quantities of data and aid decision-making has made it an integral part of modern marketing. Huang and Rust (2021) found that ML greatly increases customer engagement because it helps companies personalize their service delivery with the help of real-time consumer data. Huang and Rust (2021) differentiate between AI capability for mechanical automation, for analytical decision-making, and for comprehending human interactions. This framework is especially relevant to marketing, as different AI capabilities can be used for different purposes. Mechanical artificial intelligence can be used to automate repetitive tasks such as scheduling campaigns and basic customer service processes. Thinking AI can evaluate customer data to assist with segmentation, prediction and recommendation. More advanced artificial intelligence is able to understand language, interaction and some aspects of human emotion. Nair & Gupta (2021) found that the various ways in which company AI is applied, revealing that AI has helped them automate tasks, personalize customer outreach and support, and develop better marketing campaigns. The preceding literature review confirmed that marketers need to be more open to technologies like AI to maintain their ground in the ever-growing digital world. Nikolajeva & Teilans (2024) research accentuates the importance of machine learning in driving digital marketing personalization with a focus on an e-commerce perspective. Using unsupervised learning, the research will look at behavioral patterns and training data so it can create highly personalized advertising experiences that match your liking.

In Bangladesh, they can be applied for social media marketing, e-commerce recommendations, automated customer service, mobile financial services, and targeted advertising. But the sophistication of the implementation is likely to vary considerably between organizations. According to Davenport and Ronanki (2018), businesses can extract considerable value from AI through customer interaction, automation and insight creation, but getting it right requires the right organizational capabilities. This is important for Bangladesh as technological adoption cannot be divorced from organizational skills, data quality and infrastructure.

Omarov et al. (2022) seek to examine the adoption, use, and development of artificial intelligence in marketing through chatbots by digital marketers within the service sector. Research explores how chatbots impact consumer behavior and whether this technology is effective in increasing purchase intention. Al Khaldy et al. (2022) explores how the applications of AI and predictive analytics affect digital marketing strategies by assessing their impact on return on investment (ROI), especially in email marketing, personalization, social media optimization, and audience segmentation. According to the study, companies that implement these solutions enjoy stronger user engagement, increased conversion rates, and much higher revenues. The research uses an exploration of consumer acceptance of AI chatbots to provide a holistic view of new-age marketing efforts and showcases how businesses can capitalize on these innovations to give an engaging experience and acquire customers in the increasing rivalry. The research looks at AI tools such as machine learning, natural language processing (NLP) and predictive analytics that allow companies to process large data, automate marketing activities and customize user experiences. These technologies improve the precision of marketing efforts, by providing useful insights into customer behavior. The Technology Acceptance Model (TAM) suggested that perceived ease of use as well as perceived alternative are significant determinants representing technology adoption (Davis 1989). TAM says that regarding the use of AI for digital marketing purposes, marketers or consumers are more likely to use an AI tool just because they know how easy it is (ease of use) and they feel better in a way through its utilization, so as per them does their marketing going well. For instance, AI tools such as recommendation engines and chatbots facilitate more efficient customer experience and thus result in higher levels of user adoption and engagement (Dwvedi et al., 2021).

TAM is relevant to the goal of our study as it links the ease of use and perceived usefulness of AI to the adoption of AI by marketers as well as consumers. This is well reflected in the way algorithms execute tasks automatically or provide personalized experiences (Dwivedi et al., 2021) and obvious that ease of use and perceived usefulness are determining factors for the integration of AI in digital marketing strategies.

4.2. Digital Marketing Strategies

AI has transformed digital marketing strategies, enabling more targeted, automated and personalized campaigns. AI has found its way into two major areas: personalized marketing and automated campaign management. Rogers' Diffusion of Innovations Theory (Rogers, 2003) helps explain the adoption of AI-driven strategies. Diffusion of Innovations Theory depicts how innovations in AI technologies are embraced over the period. The theory also emphasizes communication channels, social systems, and perceived innovation characteristics such as relative advantage or compatibility in determining adoption. AI bestows different advantages in terms of better targeting and efficiency to businesses within digital marketing, thereby accelerating adoption rates (Huang & Rust, 2021). The Diffusion of Innovations Theory examines AI adoption in marketing. In this body of research paper, we focus on the enhancement made by AI-driven technologies for relative advantage and comparability under transformation in the marketing domain (Davenport & Ronanki, 2018).

4.3. Digital Marketing in Bangladesh

Rathore (2022) stated the possible impact of AI intelligence on customized and focused digital marketing methods that ultimately enhance profit-loss in conglomerates according to Rathore. The study uses empirical data from leading business firms to illustrate how AI-enabled personalization boosts customer engagement, drives more sales, and ultimately results in higher profit margins. This data only emphasizes the need for businesses to integrate AI into their marketing efforts to stay competitive and sustain long-term profitability in a shifting market. In the era of AI: (Nair & Gupta 2021)-have done excellent research on various facets and applications of Artificial Intelligence in the domain by leveraging Sophisticated Technology for Collaboration and Creativity among Marketing firms. Their study shows that faster-moving digital marketers are ill-informed of AI or have lack of effective executions to consider the role of AI in disrupting traditional marketing strategies. In addition to an age disparity between the groups using language that suggests they need help from their grandchildren to navigate life, what if it can serve as a valuable read for marketers looking to incorporate AI apps with their strategic marketing plans - driving up ROI and Business outcomes in digital marketing? As consumers are spending more time on digital platforms, digital marketing is becoming more important in Bangladesh. With the rise of social media, businesses are able to communicate directly, use influencer marketing, target their advertising and engage with customers. DataReportal's latest data shows that the number of social-media user identities in Bangladesh was about 64 million as of October 2025. High advertising reach was reported on Facebook, YouTube and TikTok (DataReportal, 2025) by the same source. The rise of digital payments also bolsters the significance of digital platforms. The 2025 payment-system report of Bangladesh Bank points to a growing ecosystem with mobile financial services, QR payments, instant transfers and digital financial infrastructure (Bangladesh Bank, 2025). These developments open up opportunities for integrated marketing. The customer can discover the product via social media, interact with the seller through a digital platform, make the payment electronically and receive after-sale service via a mobile application. This is why marketing is more and more becoming part of the broader customer journey.

4.4. Customer Value

AI plays a crucial role in enhancing customer value by improving customer engagement, satisfaction, loyalty, and perceived value. By offering personalized recommendations, automating customer service, and delivering timely content, AI enhances the overall customer experience. The Customer Value Theory provides a framework for analyzing these enhancements. In the parlance of customer value theory, this concept is illustrated by $\text{Customer Value} = \text{Benefits} - \text{Costs}$ (Woodruff 1997) AI such as predictive analytics and chatbots enable personalized content by allowing for greater efficiency in customer service, leading to perceived value when the information provided is valued (Rust & Huang, 2021). The value in this research is based on Customer Value Theory, answering how AI-driven personalization and automation lead to customers perceiving higher values. AI can improve marketing messages with increased relevance and make customer interactions easier, which could lead to directly influencing identification (i.e. must-have-ness) through increased perception value elements based on satisfaction and loyalty (Verhoef, 2021). Our study aimed at determining how AI contributes to the creation of engaging customer experiences leading to a high level of sustained engagement and loyalty with brands.

4.5. Customer Engagement, Satisfaction and Loyalty

Customer involvement is the cognitive, emotional and behavioral engagement of customers with a brand or organization. AI can help people engage through personalized content, recommendations, automated communication

and real-time interaction. Verhoef (2021) argue that customer experience is created in a series of touchpoints along the customer journey. Whig et.al (2024) examines the disruptive influence of AI-enabled innovations on service marketing to illustrate how customer engagement and experience are facilitated by technologies such as machine learning, natural language processing, and predictive analytics. Personalized communications and predicting customer needs can be automated with AI so that businesses can provide tailored on time. AI has the potential to improve these touchpoints through timely and relevant communication. There are multiple explanations for this finding in the setting of Bangladesh. Digital platforms provide businesses with the opportunity to reach customers 24/7. Social-media sites, messaging apps and mobile services can help you reach out to customers frequently. AIs can assist organizations in handling this volume of interaction more efficiently. But engagement does not equal loyalty. A customer may like, comment on or interact with a brand but not buy from it time and time again.

Customer satisfaction is the degree to which customers' expectations are met or exceeded. Artificial intelligence can boost satisfaction through faster response, personalized recommendations and more convenient service. This corresponds to the wider literature which suggests artificial intelligence may improve customer experience when technology increases relevance, convenience and responsiveness (Huang and Rust, 2021). The practical relevance for Bangladeshi companies is important. Facebook, Messenger, websites and mobile apps have created high customer expectations for fast responses. AI-enabled customer service can help organizations respond to large numbers of customers without a proportional increase in staff. However, automation is not a replacement for human help in all cases. Complex complaints, financial disputes, sensitive service issues and emotionally demanding interactions may need human intervention.

Engagement or satisfaction is more short-term than customer loyalty. It includes repeat buying, preference, commitment and willingness to keep a relationship with brand. This is of importance in theory. Thus, the benefits of AI may not always lead to continued customer loyalty. A customer may value personalized communication but still switch to another brand based on price, quality of product, availability or trust. The finding also questions overly simplistic assumptions that personalization enabled by AI automatically drives loyalty. Customer value propositions are likely to be a determinant of loyalty. Rane (2023) tries to explore how other cutting-edge technologies such as AI, IoT, and Big Data can complement each other in building customer loyalty through increased satisfaction, engagement, and relationship inexperience. The research underscores the power of IoT to capture real-time customer data and drive personalized experiences, with AI uniquely positioned to evaluate this information to understand changing customer preferences and build long-term loyalty. Further, he presents Blockchain digesting with AI and IoT to perform secured transactions so well, explaining the way technologies play a major role for businesses in deciding on how they can strategize retaining customers through innovative methodologies.

Literature has three major gaps. First, much of the AI marketing research is focused on developed markets. There is comparatively less empirical work on emerging-market environments such as Bangladesh. Second, studies tend to focus on individual outcomes such as engagement or satisfaction. Few studies are available on engagement, satisfaction, loyalty and perceived value combined. Third, the technology adoption is sometimes assumed to be good per se. The results of this study suggest a more complex interpretation: AI is highly related to engagement and satisfaction, but not necessarily to loyalty or perceived value. In the Bangladesh context, this is further complicated by the fact that the uptake of AI is taking place in a context of rapid digital growth, but also of continued digital inequality and developing institutions of AI governance.

5. Research Objectives

The objectives of this research are:

General Objective:

- To investigate the impact of AI on digital marketing strategies in Bangladesh perspectives. It examines the effect of AI on personalization, efficiency, and decisions based on customer values.

Specifics Objectives:

- AI tools (machine learning, predictive analytics, NLP) bring value to marketing processes-more targeted communications, automation of orchestration activities or delivery of personalized in Bangladesh context.
- AI-powered chatbots and recommendation engines enhance consumer engagement and satisfaction due to faster support participation & personalized interactions
- Encourages more meaningful dialog to build stronger customer relationships · Intelligent automation: reaffirming the significance of customers as valued members of the business eco-system

6. Conceptual Framework and Hypothesis Development

6.1. Conceptual Framework

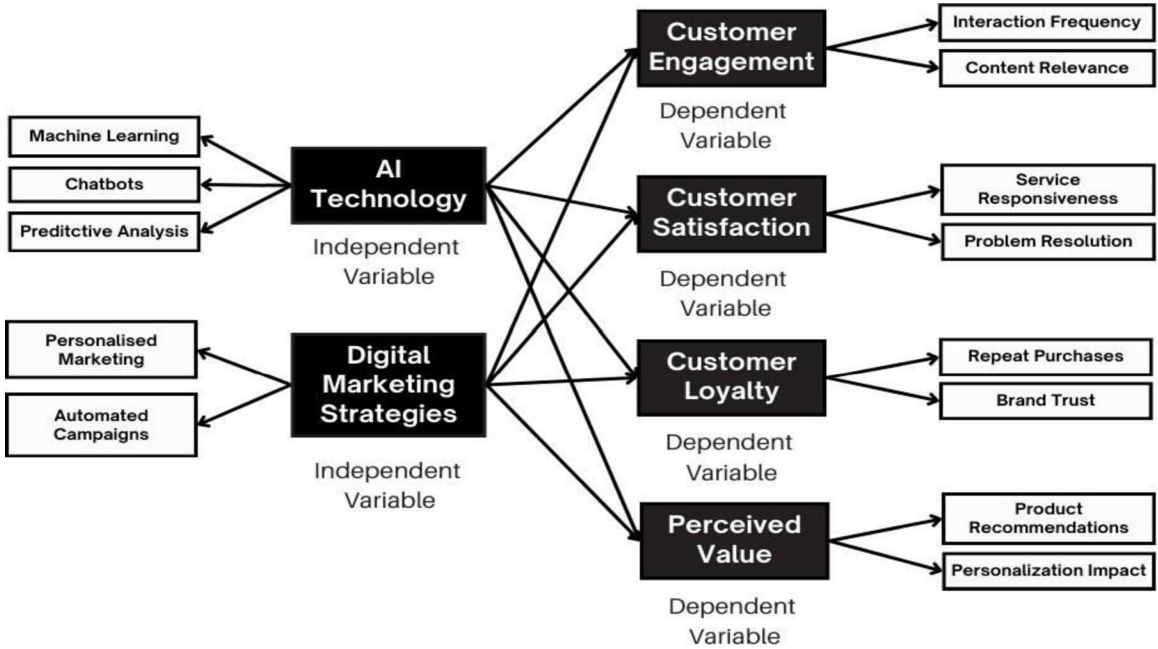


Figure: Conceptual Framework

6.2. Hypothesis Development

- *H1: The implementation of AI technologies in digital marketing significantly enhances customer engagement.*
- *H2: Personalized marketing strategies driven by AI lead to higher customer satisfaction.*
- *H3: Increased customer engagement through AI results in greater customer loyalty.*
- *H4: The use of AI in digital marketing positively influences the perceived value of products/services by customers.*

7. Research Methodology

The research projected, situated within the discipline of Artificial Intelligence (AI) and employed a mixed-methods design, aims to analyze consumer engagement behavior through digital marketing strategies using both qualitative and quantitative methods. Thus, the research behind it was based on mixed methods, quantitative and qualitative data. The quantitative part was based on a structured questionnaire and the qualitative part on interviews with relevant professionals. The empirical data set includes 167 observations. The underlying thesis is that the respondents include marketing professionals and qualified buyers from areas such as e-commerce, technology, finance and retail. Quantitative data were collected using structured questionnaires. The qualitative part comprised interviews with industry experts. The study looked at AI-associated technologies, including machine learning, natural language processing, chatbots and predictive analytics. Data was collected cross-sectionally at one point in time from participants across the demographic. This has allowed us to see in real time the impact of AI on marketing tactics and customer outcomes (engagement, satisfaction, loyalty and perceived value). The main instrument used to collect data was a questionnaire for quantitative figures. I measured satisfaction, engagement and loyalty across the specific artificial intelligence (AI) solutions' using Likert scales Open-Ended Questions: Several open-ended questions were asked to qualify into customer experience half with a focus on understanding (qualitative inclusion) a-description of consumer opinions about levels and quality sulfide trial parameters including his or her iron/output. SPSS was performed for quantitative analysis.

Principal variables were described by descriptive statistics. Bivariate relationships were evaluated using Pearson correlation analysis. The relationship between Engagement with Customers, Customer Satisfaction, Customer Loyalty and Perceived Value and Artificial Intelligence Technology was assessed by multiple regression analysis. Qualitative interview data were analyzed using thematic analysis.

8. Discussion and Findings

8.1. Descriptive findings

Descriptive info were carried out for the main variables of this research, i.e., Artificial Intelligence Technologies (AT), Digital Marketing Strategies (DMS), Customer Satisfaction (CS), Consumer Engagement (CE), Customer Loyalty (CL), and Perceived Value (PV), so as to have a general idea of the perceptions of respondents about artificial intelligence (AI) technology in digital marketing and its impact on customer value. Summary statistics for each variable are presented in Table 1 and include the mean, minimum, maximum, and standard deviation.

Descriptive Findings					
	N	Minimum	Maximum	Mean	Std. Deviation
AT	167	2.33	5.00	3.9055	.82635
DMS	167	2.00	5.00	4.0149	.98846
CS	167	2.00	5.00	4.1866	.90385
CE	167	2.00	5.00	3.9328	.96790
CL	167	1.00	5.00	3.5746	1.22243
PV	167	1.00	5.00	3.4851	1.25520
Valid N (listwise)	167				

Table-1: Descriptive Statistics of Study Variables.

The means of the variables suggest generally positive responses from the participants, indicating favorable perceptions of AI in digital marketing:

- The mean of artificial intelligence (AT) was 3.91 (SD = 0.83), indicating that respondents generally had a positive view of AI in the setting of digital marketing, with responses tending to be at a high level of agreement.
- Digital Marketing Strategies (DMS) had the highest mean score (M = 4.01, SD = 0.99), which suggests that AI-based strategies are viewed as significantly improving marketing processes such as personalization, automation and campaign management.
- Customer Satisfaction (CS) also had a high mean score (M = 4.19, SD = 0.90) meaning that most of the respondents agree that AI-driven marketing strategies improve their overall satisfaction with businesses. The Consumer Engagement (CE) variable (M = 3.93, SD = 0.96) was also positively rated, with respondents indicating that artificial intelligence (AI) tools promote higher engagement through personalized and targeted marketing content.
- The means for Customer Loyalty (CL) and Perceived Value (PV) were found to be lower at 3.57 (SD = 1.22) and 3.49 (SD = 1.26) respectively. The results suggest that AI has a positive effect on customer loyalty and perceived value, although the effects may not be as pronounced as they are for satisfaction and engagement.
- The standard deviations of Customer Loyalty (CL) and Perceived Value (PV) were relatively higher than other variables, which shows more variability in responses. This shows that some customers may believe that AI makes their loyalty and perception of value much better, while others may not think the same.
- Overall, these descriptive statistics provide a preliminary understanding of the perception of AI-driven digital marketing strategies regarding customer satisfaction, engagement, loyalty and perceived value. The mean values across variables are generally high, indicating a favorable view on the role of AI to improve marketing efforts and customer experiences. However, the greater variability in CL and PV suggests that more work is needed to explore how standard artificial intelligence can be optimized to build customer loyalty and perceived value.

8.2. Correlation Analysis

A Pearson correlation analysis, a commonly accepted methodological tool due to its reliability in investigating the interdependencies of continuous variables, was meticulously conducted to uncover the complex interrelations among the central constructs of our research, namely Artificial Intelligence Technology (AT), Digital Marketing Strategies (DMS), Customer Satisfaction (CS), Consumer Engagement (CE), Customer Loyalty (CL), and Perceived Value (PV). Table-2 shows empirical results that are evidence of the complex web of relationships that make up the digital marketing landscape.

AT			DMS	CE	CS	CL	PV
AT	Pearson Correlation	1	.908	.936	.923	-.005	.016
	Sig. (2-tailed)		.000	.000	.000	.965	.900
	N	67	67	67	67	67	67
DMS	Pearson Correlation	.908	1	.903	.934	.018	.046
	Sig. (2-tailed)	.000		.000	.000	.886	.712
	N	67	67	67	67	67	67
CE	Pearson Correlation	.936	.903	1	.918	-.044	-.010
	Sig. (2-tailed)	.000	.000		.000	.725	.935
	N	67	67	67	67	67	67
CS	Pearson Correlation	.923	.934	.918	1	-.044	-.031
	Sig. (2-tailed)	.000	.000	.000		.726	.804
	N	67	67	67	67	67	67
CL	Pearson Correlation	-.005	.018	-.044	-.044	1	.971
	Sig. (2-tailed)	.965	.886	.725	.726		.000
	N	67	67	67	67	67	67
PV	Pearson Correlation	.016	.046	-.010	-.031	.971	1
	Sig. (2-tailed)	.900	.712	.935	.804	.000	
	N	67	67	67	67	67	67

Correlation is significant at the 0.01 level (2-tailed).

Table-2: Correlation of Study Variables.

The correlation matrix, a collection of statistical data, shows some interesting relationships between the variables mentioned above, especially at the 0.01 significance level in a two tailed test, which implies a high confidence in the relationships found. The major findings are unpacked as follows:

Artificial Intelligence Technology (AT) has important positive relations with:

- Digital Marketing Strategies (DMS) ($r = .908$, $p < .01$), which indicates a strong relationship where a positive attitude toward AI is associated with a wider application of AI within the framework of digital marketing strategy.
- Consumer Engagement (CE) ($r = .936$, $p < .01$). AI is positively related to a positive perception of AI and high levels of customer involvement, confirming AI's role in enabling dynamic interactions with customers.
- Customer Satisfaction (CS) ($r = .923$, $p < .01$) suggests a significant association between a positive attitude toward AI and higher customer satisfaction, a key metric for measuring marketing effectiveness.

Digital Marketing Strategies (DMS) are highly correlated with:

- Client Engagement (CE) ($r = .903, p < .01$), indicating that the strategic application of AI-enhanced marketing techniques is necessary for enhancing consumer engagement, thus highlighting the necessity to adopt such cutting-edge approaches.
- Customer Satisfaction (CS) ($r = .934, p < .01$) This supports the hypothesis that the use of AI in marketing strategies is likely to result in higher satisfaction levels among the consumer base.
- Consumer Engagement (CE) and Customer Satisfaction (CS): The relationship between CE and CS was highly significant ($r = .918, p < .01$) which means more engagement created by AI interventions has a significant positive effect on overall customer satisfaction.
- In contrast, Customer Loyalty (CL) and Perceived Value (PV) do not show significant correlations with the other variables which are Artificial Intelligence Technology (AT) and Digital Marketing Strategies
- Customer Satisfaction (CS) and Client Engagement (CE) (DMS). This salient observation indicates that these two constructs are closely interlinked with each other ($r = .971, p < .01$) although they do not originate directly from AI related factors. This relation shows that perceived value is a significant antecedent of customer loyalty, an important aspect of sustainable brand success.

Key Insights:

- Substance Correlations Between AI Perception and Marketing Effectiveness The empirical analysis shows strong correlations between the implementation of AI technologies, the effectiveness of digital marketing paradigms, the level of consumer engagement, and the level of customer satisfaction. This intricate interplay indicates a causal nexus, where positive perceptions of AI in marketing are contiguous with better customer-centric outcomes. Organizations adopting AI-infused marketing strategies are probably better placed to experience increased engagement and resulting satisfaction amongst their customers.
- Perceived Value is an Indirect Medium of AI's Effect on Customer Loyalty: There is a strong association between Customer Loyalty and Perceived value, which indicates the importance of the latter in influencing the loyalty patterns even when there is no statistically significant direct link between AI-related constructs and customer loyalty. This insight suggests that commercial entities should strategically focus on using AI to enhance the perceived value proposition and act as a catalyst for developing long-term customer relationships and loyalty.

8.3. Regression analysis

The model summary of variables in Table-3 presents the regression analysis results containing the coefficients of determination and the fit of the model. R-value of .950 reflects the presence of a strong positive correlation between independent variables (CE, CS, CL, PV) and the dependent variable (AT) that is statistically significant. The value of R Square equal to .903 suggests that 90.3% of the variance in Attitude toward AI (AT) can be explained by the interaction of the customer involvement (CE), Customer Satisfaction (CS), Customer Loyalty (CL), and Perceived Value (PV). A high R-square value demonstrates the explanatory power of the model.

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.950 ^a	.903	.897	.26509
a. Predictors: (Constant), PV, CE, CS, CL				

Table-3: Model summary of Study Variables.

The Adjusted R Square equal to .897 is used to improve the prediction model's accuracy. Thus, accounting for the possible inflation caused by the presence of multiple independent variables to achieve better goodness-of-fit. In this way, the accuracy of prediction model is improved to have high goodness of fit in spite of the complexity of analysis.

Model	Squares	Sum of	df	Mean	F	Sig.
				Square		
1	Regression	40.711	4	10.178	144.827	.000
	Residual	4.357	62	.070		
	Total	45.068	66			

Dependent Variable: AT

Predictors: (Constant), PV, CE, CS, CL

Table-4: ANOVA

The Analysis of Variance (ANOVA) in Table-4 gives a general picture of the significance of the model. The F statistics of 144.827 and significance level of $p < .001$ results indicate that the independent variables have a significant impact on artificial intelligence (AT) simultaneously. This confirms the usefulness of the model in showing the basic processes that influence consumer opinions on AI in digital marketing.

Model		Unstandardized		Standardized	t	Sig.
		Coefficients		Coefficients Beta		
		B	Std. Error			
1	(Constant)	.354	.186		1.903	.062
	CE	.485	.088	.566	5.532	.000
	CS	.371	.093	.406	3.998	.000
	CL	.049	.115	.072	.425	.673
	PV	-.024	.112	-.036	-.212	.833

Dependent Variable: AT

Table-5: Coefficients

The unstandardized coefficients (B) measure how strong the link is between the predictor variables and the outcome of interest Attitude toward Artificial Intelligence. Table-5 shows that Client Engagement is significantly and positively related to Attitude toward Artificial Intelligence ($B = .485$ $p < .001$). The coefficient is large, meaning that for each one-unit rise in Client Engagement, Attitude toward Artificial Intelligence is expected to rise by .485 units provided all other variables stay the same. The Beta coefficient is .566 which highlights the impact of Client Engagement on Attitude toward Artificial Intelligence. Client Engagement is the influential determinant in the analysis model. Customer Satisfaction is also a factor. It has a positive effect on Attitude toward Artificial Intelligence ($B = .371$ $p < .001$). The Beta coefficient of .406 supports the claim that Customer Satisfaction's a very influential variable in forming Attitude toward Artificial Intelligence and second only to Client Engagement in predictive power. In contrast Customer Loyalty shows a coefficient of .049 with a p-value of .673 indicating it is not statistically significant. This demonstrates that Customer Loyalty is part of the broader marketing discussion but does not strongly influence how people view Attitude toward Artificial Intelligence. The study on Perceived Value shows a result for its predictive power on Attitude toward Artificial Intelligence ($B = -.024$ $p = .833$). The small negative coefficient indicates no link between Attitude toward Artificial Intelligence and Perceived Value. In terms the coefficient analysis reveals how client engagement, customer satisfaction, customer loyalty and perceived value relate to Attitude toward Artificial Intelligence. The significant results for Client Engagement and Customer Satisfaction underscore their importance in shaping Attitude toward Artificial Intelligence. The non-significant results for Customer Loyalty and Perceived Value suggest that these variables play a role, in the current understanding of how customers interact with Artificial Intelligence.

Key Insights:

- Customer involvement is found to be a salient antecedent in the formation of positive attitudes toward artificial intelligence (AI) in a commercial context, with a significant effect as evidenced in empirical analysis. It highlights the necessity of business organizations to prioritize the optimization of engagement mechanisms in AI-embedded marketing paradigms to promote a more favorable reception of these technologies by their clientele.

- Customer loyalty and perceived value are recognized as important determinants for the overall success of an enterprise, but they have a relatively small impact on consumers' attitudes toward AI. This observation points to a more complex role for AI in promoting loyalty and influencing value perceptions, with its effects likely to be significant in ancillary areas such as customer retention, but not necessarily in the way in which it shapes the emotional disposition toward the technology itself.

8.4. Consequence of Hypothesis

Hypothesis-1 (H1): The implementation of AI technologies in digital marketing significantly enhances customer engagement.

Outcomes:

- Customer involvement (CE) is strongly and positively linked with intelligence (AI) technologies, especially with AT (Artificial Intelligence Technology in marketing). The Pearson correlation stands at 0.936 with a p-value than 0.001 showing a very strong and statistically significant relationship. In the regression analysis CE has a positive coefficient ($B = 0.485$, $p < 0.001$) which means that artificial intelligence (AI) technologies play a major role in boosting consumer engagement.
- Effect: Accepted – There is strong evidence that AI technologies significantly enhance customer engagement.

Hypothesis 2 (H2): Personalized marketing strategies driven by AI lead to higher customer satisfaction.

Outcomes:

- Customer Satisfaction (CS) is also positively and strongly connected to intelligence (AI) technologies (AT) with a Pearson correlation of 0.923 and a p-value below 0.001. This indicates an meaningful link. The regression analysis confirms this showing a positive coefficient for CS ($B = 0.371$, $p < 0.001$). This suggests that AI-driven personalized marketing strategies lead to levels of customer satisfaction.
- Effect: Accepted – The data supports that personalized marketing strategies via AI lead to higher customer satisfaction.

Hypothesis 3 (H3): Increased customer engagement through AI results in greater customer loyalty.

Outcomes:

- However, customer loyalty (CL) shows no significant connection with artificial intelligence (AI) technologies (AT). The Pearson correlation is -0.005 with a p-value of 0.965 which's not statistically significant. In the regression analysis the coefficient for CL is not $B = 0.049$, $p = 0.673$) meaning there is no meaningful relationship between customer involvement through AI and customer loyalty in this study.
- Effect: Rejected – There is no significant evidence to suggest that increased customer engagement through AI results in greater customer loyalty.

Hypothesis 4 (H4): The use of AI in digital marketing positively influences the perceived value of products/services by customers.

Outcomes:

- Perceived Value (PV) is weakly or insignificantly correlated with intelligence (AI) technologies (AT) with a Pearson correlation of 0.016 and a p-value of 0.900. The regression analysis shows a insignificant coefficient, for PV ($B = -0.024$, $p = 0.833$) which indicates that AI does not significantly influence the perceived value of products or services.
- Effect: Rejected – The use of AI in digital marketing does not significantly influence perceived value, based on the data analyzed.

7. Conclusion

The emergence of intelligence technologies such as machine learning, natural language processing and predictive analytics has enabled businesses to analyze massive amounts of data quickly and accurately. This has resulted in a shift in marketing. Now, companies can develop marketing efforts that are precisely tailored to individual customer needs and behaviors rather than one-size-fits-all approaches. Thanks to AI's ability to analyze data in time, marketers no longer must rely on broad generic campaigns. Now, focused, situationally aware promotions are being built that

resonate more strongly with customers. Artificial Intelligence improves customer interaction and satisfaction, as demonstrated by tools such as chatbots that provide customer service 24/7 and recommendation engines that suggest products based on user habits. AI has found its way into modern marketing. It provides opportunities to improve marketing strategies, increase engagement and deliver personalized experiences to customers. Ethical concerns and practical difficulties remain, but the impact of AI on marketing is clear and undeniable. Companies that know how to use AI thoughtfully and well in their marketing activities will have an advantage over the competition. These companies will be able to deliver experiences that are more relevant, faster and more satisfying for consumers. Moving forward, it's important that organizations stay vigilant, flexible and committed to standards in how they use AI. This research has provided a basis for understanding the role of AI in digital marketing and customer value creation. This research gives some insight that can help businesses move forward in a way. Through careful informed decisions, companies will truly gain the benefits of AI. This will enable them to keep up with rising customer expectations and remain competitive and successful in a world where artificial intelligence is the innovation engine.

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